

Mill Street Labs

We envision frictionless intelligence transfer and accumulation among people and agents. We are building long-running agent populations that discover, share, and build on useful knowledge.

TEAM

Team of 4, Harvard Class of '27

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THESIS

Long-running agent populations can improve collectively through adaptive social dynamics and intelligence transfer.

Human innovation is an emergent property of long-running social dynamics; new ideas build on knowledge accumulated over generations through teaching, cooperation, and competition between groups.

- **Adaptive social dynamics:** have groups compete for finite resources, so that helping one's group is rewarded even if doing so costs an agent resources. Let agents revise how they cooperate, enter and leave swarms, and migrate between groups.
- **Intelligence transfer:** credit agents for the improvement of those they teach; use teaching to combine discoveries from specialized lineages; accrue an interpretable corpus of verified communal knowledge with easy deposit and retrieval

RESEARCH

Our focus is swarm improvement itself.

Agent swarms have spontaneously demonstrated social dynamics, but their social development is limited by short-horizon objectives, siloed reasoning, and fixed search space. We let populations run continually while perturbing organization and incentives, evaluating the impact on what agents discover and retain, to identify what sustains collective improvement.

SANDBOXES

Sandbox 1: strategy development in skillful games

Skillful games (e.g. predator-prey models, Flag Game, Werewolf, Hanabi) provide clear credit assignment, counterfactuals, and metrics for agent interaction.

- developing custom policies, corpus, gestation, and fine-tuned personalities in SmolLM2 and GPT-4o
- accelerating SmolLM2 discovery fivefold in predator-prey environment over equivalent model calls
- saturating multiple skillful games with $\leq 1/10$ th the compute of baseline models

Sandbox 2: autonomous research in financial markets

Markets let swarms explore, build large ideas, innately reward ingenuity, and adapt to dynamic multi-variate settings. We have domain knowledge to evaluate valuable problems and candidate strategies.

- 3–4 ms roundtrip execution with the Kalshi matching engine
- deployable positive expectancy strategies for market making